PEI YU CHANG

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SUMMARY

Ph.D. candidate in Mechanical Engineering at The Ohio State University Center for Automotive Research (CAR), graduating May 2027, with research in risk-aware and safe control for autonomous vehicles. 9+ years of engineering experience, including systems and project engineering at NCSIST on UAV and naval systems, an Autonomy Capabilities Systems Engineering internship at Motional, and manufacturing automation at TSMC. Experienced in systems engineering, verification and validation, and controls for autonomous vehicles and robotics, with hands-on work from requirements and test planning through simulation, dSPACE HIL, and full-scale vehicle testing.

EDUCATION

The Ohio State University, Columbus, OH
Ph.D. Mechanical Engineering | GPA: 3.9/4 — Expected May 2027
M.S. Mechanical Engineering | GPA: 3.9/4 — May 2026
Center for Automotive Research (CAR): Risk-aware and safe control for autonomous vehicles under GPS uncertainty and multimodal sensor disturbances, applied to collision avoidance and cooperative driving.

National Cheng Kung University, Tainan, Taiwan
M.S. System and Naval Mechatronic Engineering | GPA: 3.6/4 — Aug. 2016

SKILLS

- **Systems Engineering:** Requirements development & traceability, system architecture & design, interface definition, hardware/software integration, cross-functional collaboration
- **Test & Validation:** Requirements-based V&V, test planning & execution, scenario-based testing, test case development, pass/fail criteria, SIL/HIL, full-scale vehicle testing, test metrics, data/log analysis, root-cause troubleshooting, FAT/SAT
- **Vehicle Systems & Controls:** ADAS, autonomous vehicle testing, vehicle control systems, safety-critical control (MPC, CBF), CAN (CANalyzer), Ethernet/TCP-IP, sensor integration
- **AI & Computing:** End-to-End Autonomous Driving, Reinforcement Learning (PPO, SAC), World Models, Physical AI, Agentic AI, HPC, Slurm job scheduling, Linux, test automation
- **Software & Tools:** Python, C++, MATLAB/Simulink, dSPACE SCALEXIO, CANalyzer, ROS2, Git/GitHub, LabVIEW, Applied Intuition

ENGINEERING EXPERIENCE

Motional, Pittsburgh, PA — **May 2025 – Aug. 2025**
Intern, Autonomy Capabilities Systems Engineering

- Developed 13 requirements-based validation scenarios for safety-critical emergency vehicle interactions, translating expected system behaviors into test conditions, verification criteria, and evaluation metrics.
- Designed system-level validation metrics using relative kinematics, time-to-collision, and interaction context to identify safety-critical agents and potential collision scenarios from real-world fleet data.
- Evaluated validation metrics across diverse real-world driving scenarios, achieving 96% recall with a 2% false-positive rate to support data-driven system safety assessments.

CARMEN+, U.S. Department of Transportation (DOT), Columbus, OH — **2023 – Present**
OSU Graduate Research Assistant

- Developed safety-critical autonomous driving functions using Model Predictive Control (MPC) and Control Barrier Functions (CBFs) for collision avoidance and multi-vehicle cooperative control.
- Planned and executed simulation, dSPACE HIL, and full-scale vehicle testing to validate control and collision avoidance functions under dynamic traffic, sensor uncertainty, and V2V communication scenarios.
- Conducted vehicle-level validation on a GM Bolt at TRC SMARTCenter, evaluating controller performance, safety responses, and consistency between simulation, HIL, and real-world testing.
- Developed Python-based testing and analysis workflows integrated with dSPACE to automate experiments, analyze system responses, and evaluate autonomous control and reinforcement learning policies (PPO/SAC).

National Chung-Shan Inst. of Sci. & Tech, Taoyuan, Taiwan — **Nov. 2016 – May 2018, Oct. 2019 – July 2023**
System Engineer / Project Engineer

- Managed requirements traceability across customer specifications, system design, supplier deliverables, and verification activities, and supported system design reviews for mission-critical naval systems.
- Led system integration and communication interface design for a 20-UAV autonomous fleet, developing system requirements and verification plans to support three successful missions.

- Redesigned a legacy Ground Control Station (GCS) architecture, integrating communication networks, subsystem interfaces, and system redundancy to improve operational reliability.
- Coordinated suppliers, shipyard, and engineering teams through hardware/software integration, troubleshooting, functional verification, and factory acceptance testing (FAT) for field deployment.

TSMC (Taiwan Semiconductor Manufacturing Company), Tainan, Taiwan **May 2018 – July 2019**

Team Leader, Manufacturing Section

- Improved manufacturing tool productivity by 30% by leading automation and process improvement initiatives, including cleanroom AGV deployment and manufacturing data system enhancements, in 24/7 operations.

SELECTED ENGINEERING PROJECTS

Buckeye AutoDrive, GM-Sponsored Autonomous Vehicle Competition **2023-2025**

- Integrated and troubleshot vehicle control interfaces with perception and planning subsystems, using CANalyzer for CAN communication monitoring and debugging.
- Developed and deployed Pure Pursuit and MPC path-tracking controllers, and demonstrated autonomous fishbone parking on a full-scale vehicle through controller integration, tuning, and real-world testing.

Transit Collision Avoidance & Braking Notification, Federal Transit Administration (FTA) **2026-Present**

- Developed the system concept and proposed architecture for a transit bus collision avoidance and braking notification system, defining system functions and technical requirements.
- Led technical planning for the $650K, two-year FTA-funded project, translating project objectives into engineering tasks, system development activities, and validation plans.

AI-Powered Mobile Infrastructure for Road Safety, USDOT SS4A ($5M) **2026-Present**

- Contributed to a system concept using transit buses as mobile AI-powered infrastructure for road safety monitoring and collision prevention, and supported system architecture and technical planning.

HPC Research Computing & Workflow Automation, Ohio Supercomputer Center (OSC) **2023-Present**

- Developed Python-based research workflows for experiment orchestration, code review, and result reporting, running validation workloads on Linux-based HPC resources with Slurm job scheduling.

PUBLICATIONS (FIRST AUTHOR)

- "Barrier Function Conformal Safety Clearance Certification with CVaR for Driving Trajectory Selection," IEEE Robotics and Automation Letters (RA-L), Under Review, 2026.
- "Risk Aware Safe Control with Cooperative Sensing for Dynamic Obstacle Avoidance," Control Engineering Practice, 2026, presented at the 23rd IFAC World Congress.
- "Risk-Budgeted Control Framework for Balanced Performance and Safety in Autonomous Vehicles," 2026 IEEE American Control Conference (ACC).
- "Experimental Evaluation of Model Predictive Control and Control Barrier Functions for Autonomous Driving Obstacle Avoidance," 2025 IFAC 13th Symposium on Nonlinear Control Systems (NOLCOS).
- "Enhancing Safety at Highway Ramps and Intersections Using Safe Cooperative Controls in Automated Vehicles," 2024 IEEE International Conference on Intelligent Transportation Systems (ITSC).

PRESENTATIONS

- "Centralized Policy Learning for Consensus Control of Connected and Automated Vehicles," AAAI Multi-Agent AI in the Real World Workshop, 2025.
- "Optimizing Safety at Highway Ramps and Intersections: Implementing Safe Cooperative Control Systems in Automated Vehicles," CCAT Global Symposium on Mobility Innovation, 2024. Poster Presentation.
- "Building Safer Communities with Next-Generation Autonomous Vehicle Technologies," Engaged Scholarship Symposium – Graduate Student Poster Competition, 2024. Poster Presentation.

HONORS & SERVICE

- **Awards:** American Control Conference (ACC) Student Travel Grant, 2026; IEEE ITSC Student Travel Grant, 2024
- **Peer Reviewer:** Reviewed 9 manuscripts for the ASME Journal of Dynamic Systems, Measurement, and Control, Control Engineering Practice, ACC, IFAC conferences, and IEEE ITSC.